

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and Environmental Engineering) SEMESTER - 4 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and Environmental Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTCEC405 - WASTE WATER ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 The average quantity of wastewater generated is generally taken as:
 - a) 50% of water supplied
 - b) 60% of water supplied
 - c) 80% of water supplied
 - d) 100% of water supplied
 - 2 Grit chambers are designed to remove particles having size greater than:
 - a) 0.05 mm
 - b) 0.10 mm
 - c) 0.20 mm
 - d) 0.30 mm
 - 3 Which of the following is a **suspended growth process**?
 - a) Trickling filter
 - b) Rotating biological contactor
 - c) Activated sludge process
 - d) Bio-tower
 - 4 Anaerobic treatment is most suitable for wastewater having:
 - a) Low BOD
 - b) High BOD
 - c) High dissolved oxygen
 - d) High nitrate content
 - 5 Septic tanks are commonly used for:
 - a) Industrial effluents
 - b) Large municipal sewage
 - c) Small community wastewater
 - d) Storm water treatment
 - 6 Conditioning of sludge is done before:
 - a) Thickening
 - b) Stabilization
 - c) Dewatering
 - d) Disposal

- 7 The DO sag curve represents the variation of:
 - a) BOD with distance
 - b) DO with time or distance
 - c) pH with depth
 - d) Velocity with discharge
- 8 The main purpose of disinfection is to:
 - a) Remove organic matter
 - b) Improve colour and taste
 - c) Destroy pathogenic organisms
 - d) Reduce turbidity
- 9 Streeter–Phelps model is used to predict:
 - a) Sedimentation efficiency
 - b) Sludge digestion rate
 - c) Oxygen deficit in streams
 - d) Effluent filtration efficiency
- 10 Sludge recycled in the activated sludge process is known as:
 - a) Primary sludge
 - b) Waste activated sludge
 - c) Return activated sludge
 - d) Digested sludge
- 11 Trickling filters are classified as:
 - a) Suspended growth systems
 - b) Anaerobic systems
 - c) Attached growth systems
 - d) Chemical treatment systems
- 12 Peak factor in sewage pumping is used to account for:
 - a) Population growth
 - b) Variations in flow rate
 - c) Pump efficiency losses
 - d) Sewer leakage

Q2. Solve the following.

- A) Summarize the significance of self-cleansing velocity in sewer design. 6
- B) A city with a population of 100,000 has an area of 50km². Rate of water supply is 110 litres per capita per day of which 80% turns into sewage. The average run-off coefficient is 0.5 and intensity of rainfall is 14.5mm/hr. Estimate the quantity of combined sewage. Take peak factor as 2.5. 6

Q3. Solve the following.

- A) Explain the terms BOD and COD. Differentiate between first and second stage BOD. 6
- B) B.O.D. of a sewage incubated for one day at 30°C has been found to be 120 mg/l. What will be its 5 day 20°C BOD, if K at 30°C is 0.16 per day (base 10). 6

Q4. Solve Any Two of the following.

- A) Discuss the choice available and the factors to be considered while selecting pumps and pipes for sewerage system and explain. 6
- B) Design a circular primary sedimentation tank to treat an average sewage flow of 5000 m³/day, suitably assuming the design criteria. Draw a neat sketch of the designed tank. 6

- C) List the objective and functioning principle of a secondary treatment unit. 6
- Q5. Solve Any Two of the following.
- A) Write in detail about the UASB reactor with neat sketch, advantages and disadvantages. 6
- B) Explain the various process involved in sludge treatment and disposal. 6
- C) State the Indian standards for sewage disposal on land and conditions favoring it. 6
- Q6. Solve Any Two of the following.
- A) Describe with neat sketches about the trickling filters. 6
- B) Explain the two types of sewage system. 6
- C) Discuss the operational principles of activated sludge process. 6

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